

Feasibility and optimal operation of micro energy hybrid system (hydro/wind) in the rural valley region

Nesrine Kefif¹, Bachir Melzi², Mehran Hashemian^{3,*}, Mamdouh El Haj Assad⁴ and Siamak Hoseinzadeh⁵

¹Electrical and Electronic Institute, University of Boumerdes, Road of 1^{er} November, CP 35063, Boumerdes, Algeria; ²State Key Laboratory of Advanced Technology for Materials Synthesis and Processing, School of Materials Sciences and Engineering, Wuhan University of Technology, 122 Luoshi Road, Hongshan District, CP 430070, Wuhan, Hubei, China; ³Department of Mechanical Engineering, Faculty of Engineering, Urmia University, 11km SERO Road, CP 5756151818, Urmia, Iran; ⁴Sustainable and Renewable Energy Engineering Department, University of Sharjah, Sharjah, University City PO Box 26666, Sharjah, United Arab Emirates; ⁵Department of Planning, Design, and Technology of Architecture, Sapienza University of Rome, Via Flaminia 72, 00196 Rome, Italy

Abstract

With the increase in the global and local demand for electrical energy, which is necessary for the functioning of several fields such as the economy and agriculture, this study introduces a micro-hydro-wind hybrid system in order to obtain an economic feasibility of the off-grid isolated and renewable energy system. The main objective of this research is to determine the optimum design size for a hydro-wind hybrid energy system that is supposed to meet the demand for the electric load in front of a valley in Algeria called Yesser with variable water flow and wind speeds in three positions. Batteries are supplied to increase the reliability of the system when the performance of the wind turbine energy and the flow rate of the hydro system are evaluated. This system is proposed to reduce financial costs in addition to the possibility of providing interchangeable energy and operating reserves with short start times. The Homer Pro software is used to model the hybrid renewable energy system and to perform the required analysis of the economic side of the system in terms of the valley's flow rate and the wind speed. The average speed of the water flow in Yesser valley is varied between 10 and 24.6 m³/s, and three cases of the valley's flow rate were studied with the maximum and minimum wind speeds: the passage (the transit) with 10 m³/s, the downstream (the estuary) with 19 m³/s and the valley's upstream (the source) with 24.6 m³/s. From the results, it appears that the hydro power was not enough to meet the load demand in the first area (the transit); however, the hydro/wind hybrid system was capable to feed the required load. For the second area (the estuary) the wind power was needed just in summer season, and for the third area (the source) the hydro generator was sufficient to feed the load all the year as the hydro power generated exceeds the load demand. The Homer Pro software calculations prove that the financial cost of the system is inversely proportional with the wind speed, such that the higher the wind speed the lower the cost of the system, the same relationship is for the flow rate speed as we consider the third case the most cost-effective with 61 330.46 USD.

Keywords: financial cost; HOMER; hydropower; off-grid; wind energy

*Corresponding author:

m.hashemian@urmia.ac.ir;

hashemian.mehran@yahoo.com Received 16 July 2021; revised 14 September 2021; accepted 29 September 2021

1 INTRODUCTION

The first electricity generation was based on fossil fuel burning. The gas emissions from the fossil fuel burning is the main cause of global warming [1, 2], which is the reason that many researchers have shown the potential of renewable energy systems [3, 4] as a suitable replace for diesel generators, where the wind energy is the most abundant compared with the other renewable sources [5, 6]. Moreover, the practicality of hybrid systems was explored by ferrite and witness [6–9]. This paper presents an economic study of a micro-hydro–wind off-grid isolated hybrid system. Although the renewable energies have proved feasibility of themselves in off-grid systems, using single energy source leads to the large size of components and the high cost of system [10]. The huge costs of the system can be avoided by hybrid systems as each energy source complements the other [11–13]. The Merriam–Webster Dictionary describes efficiency as ‘the ratio of useful energy to energy supplied’ [14]. However, power system’s efficiency has been based on economic measures [15].

The aim of this research is to construct a design of a hybrid energy system that confirms its ability to feed the load with minimum cost. Batteries were supplied to increase the reliability of the system as the wind turbine energy and the flow speed of the hydro system were taken into account [16]. This system was proposed to provide efficient power, which helps to reduce the dependence on the electricity network and its financial costs [17–20], in addition to the possibility of providing interchangeable energy and operating reserves with short start times.

Hydropower plant is powered by downhill water flowing from the reservoirs. When there is an energy surplus, the hydropower will return the water back up the hill to the reservoirs, where it will act as a giant battery. Thus, the hydro power plant will be able to generate power when the prices are high and use power when they are low, complementing the grid as well as the wind turbines. Thanks to the complementary role of both the wind turbines and the hydro turbines, the hybrid system is able to produce power constantly with optimum cost. This system is so suitable for rural areas especially for irrigation purpose where there is no access to electricity grid [21]. The procedure of an optimal size design is described by Maheshwari and Ramakumar in [22].

Many researches in the literature focuses on hybrid systems: in a case study in the island of El Hierro (Canary Islands, Spain) the author designed a wind-hydro system to provide electricity for a population of 10,000 while ensuring the network stability [23]. Li et al. [24] designed a new hydro–wind hybrid model using MATLAB/Simulink with high energy efficiency based on the limitation of economic benefits. Eliseu and Castro [25] studied two different wind-hydro hybrid park operating strategies in order to maximize the market profit and to generate a constant output power using Artificial Neural Networks by anticipating the market prices and the wind speed. In another study, Xie et al. [26] predicted the hydro–wind power system lifetime depending on the frequency change using two capacity power ratios. Lastly, a comparative study of water storage capacity between conventional reservoir

and pumped storage plant presents results in a wind-hydro hybrid system based on Homer software [27, 28].

The objective of this research paper is the design of a wind-hydro hybrid energy system in Yesser valley for the electricity consumption to be based on green energy resources with a guarantee of energy availability throughout the year. The system is

- environmentally friendly;
- efficient enough for feeding the load;
- applying sustainable and renewable energy sources; and
- with no environmental effects.

For this purpose, the mentioned system is studied in three positions of the valley. Several tests for natural changes such as flow velocity, wind speed and flow rate have been carried out and the results have been described. The Homer Pro software [28] is used to model the Hybrid Renewable Energy System and to perform the required analysis to assist the system designer to choose the optimal design for components of the hybrid system. The best design is the combined wind-hydro power plant studied in this paper. To fulfill this aim, the climate data are downloaded from NASA website [29] and used for power output and cost calculations.

2 REGION DESCRIPTION: ‘OUED YESSER’

The Yesser valley is among the most famous North African valleys leaking into the Mediterranean basin and is one of the most important regions in the Algerian coast from flow rate viewpoints. The length of Yesser Valley is ~100 km and descends from the Tablat region in Medea province, passing through the western side of Bouira province and then to the eastern coast of Boumerdes province. Finally, it pours into the Mediterranean Sea (Figure 4). Yesser valley occupies a surface of 3615 km² (Figure 1). The study area is a semi-arid, Mediterranean climate [30]. Yesser valley is characterized by non-interruption and instability of the flow during the year in a way that corresponds to the low level of rainfall during the warm seasons. Since precipitations are more abundant in the winter than the other seasons, several farms are located on the banks of this valley that suffer from multiple power interruptions. Hence, this area is more suitable for the construction of this project. In order to conduct the overall study of all cases of the valley, three different areas of Yesser valley are carried out.

The information was taken from the world weather database [31] and NASA weather database [29].

- The first case (area 1 in Bouira province) transit: passage:

It is a semi-flat steppe zone, which is one of the branches of the Yesser valley and is surrounded by some agricultural farms. The yearly average wind speed ranges between 1.8 and 4.69 m/s, and the water flow rate of the valley in this region ranges between 8 and 12 m³/s, with an average of 10 m³/s. The load in this area is estimated to be 10.934 kW. Figure 1 shows the monthly wind speed in Bouira province for 1 year.

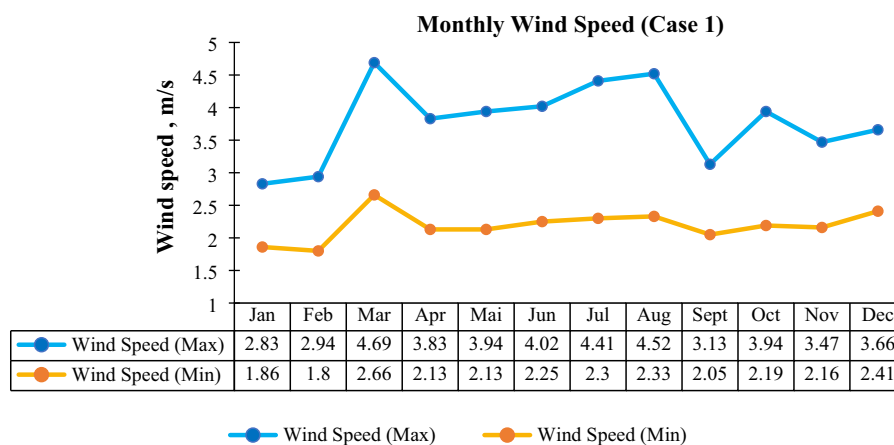


Figure 1. Monthly wind speed in Bouira province.

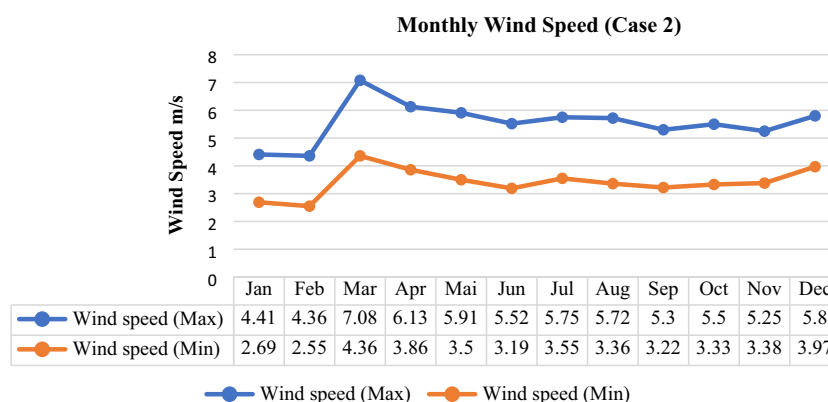


Figure 2. Monthly wind speed in Boumerdes province.

- The second case (area 2 in Boumerdes province) downstream: the estuary:

This is a coastal area, and there is a small coastal village with an average wind speed ranging from 2.55 to 7.08 m/s and the average valley water flow rate in this village is $\sim 19 \text{ m}^3/\text{s}$. The load in this area is estimated to be 9.254 kW.

Figure 2 shows the monthly wind speed in Boumerdes province for 1 year.

- The third case (area 3 in Media province) upstream: the source:

It is a lower sloping area surrounded by mountains (1810 m), but the flow rate in this area is very fast and reaches $25.3 \text{ m}^3/\text{s}$. The load in this area is estimated to be 11.465 kW.

Figure 3 shows the monthly wind speed in Medea province for 1 year.

3 POWER SYSTEM DESCRIPTION

Figure 5 shows a simplified diagram of the micro-hydro–wind hybrid power system.

3.1 Hydro turbine

The initial capital cost for a hydro turbine is 50 000 USD, which includes some civil works. Its replacement is priced at 40 000 USD. The annual operation and maintenance cost is 1 000 USD. The hydro turbine is estimated to bear a lifetime of 30 years. The turbine has a nominal power of 16.554 kW. Table 1 shows the monthly steam flow in the principal side of Yesser valley.

3.2 Wind turbine

The wind turbine has a capacity of 10 kW. Its initial cost is 50 000 USD and its replacement is at 50 000 USD. Annual operation and maintenance cost is 500 USD. Its hub and anemometer are located at a height of 30 m. The wind turbine is estimated to bear a lifetime of 20 years. The monthly wind speed is as shown in Table 2, which was taken from the Homer modeling software [28].

Table 2 also shows the wind turbine power curve in terms of wind speed: the highest value of wind turbine power output is 10 kW at wind speeds between 14 and 15 m/s and the lower values less than 2 kW before reaching 7 m/s wind speeds. Table 3 shows the maximum and minimum wind speeds for each study case.

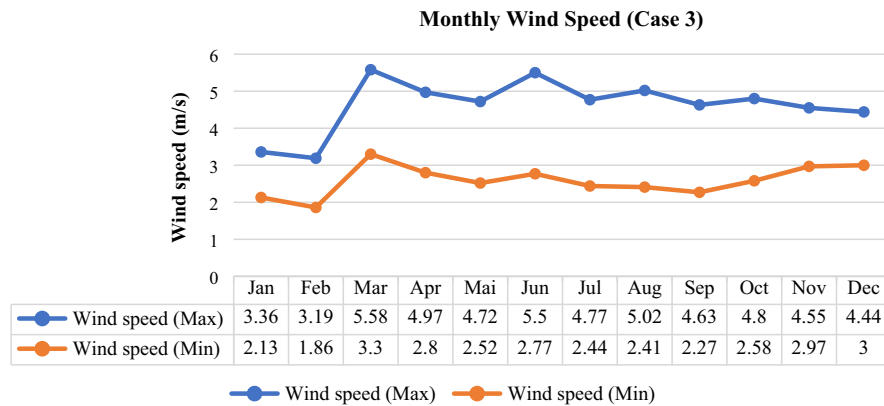


Figure 3. Monthly wind speed in Media province.

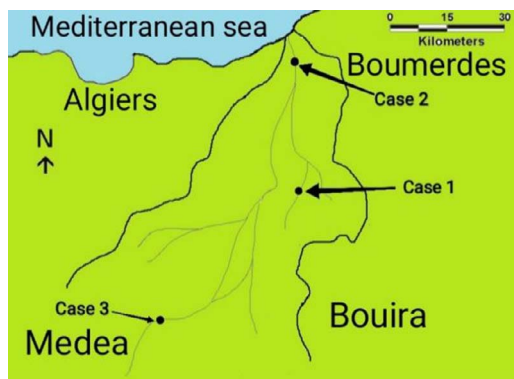


Figure 4. Localization of different cases (area) of Yesser valley.

Table 1. Monthly stream flow.

Month	Stream flow (m ³ /s)
January	29
February	31
Mar	30
April	30
May	28
June	25
July	22
August	16
September	22
October	20
November	23
December	26

Table 2. Wind speed versus turbine power output.

Wind speed (m/s)	Power output (kW)
0	0
4	0.19
5	0.37
6	0.93
7	1.85
8	3.33
9	5.19
9	7.04
10	8.52
12	9.44
13	9.81
14	10
15	10

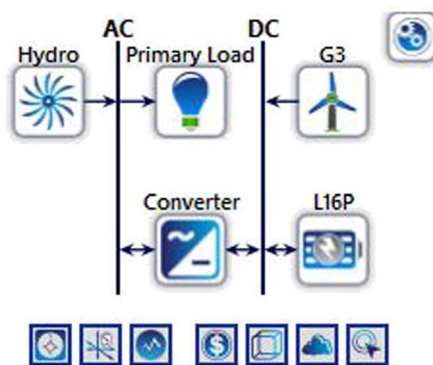


Figure 5. Simplified diagram of the hydro-wind hybrid power system.

3.3 The battery

The valve-regulated lead-acid battery rate is at 6 V and has a capacity of 396 Ah. Twelve batteries initially cost 210 USD. The replacement batteries will cost another 210 USD. The operation and maintenance cost add further 4 USD to the annual total cost. The general information of battery is provided in Table 4.

3.4 Converter

The converter has a capacity of 5 kW. Its initial capital cost is 200 USD and its replacement is at 20 USD. The operation and maintenance cost add further 10 USD to the annual total cost. The lifetime and efficiency of inverter input were considered at 20 years and 90% efficiency, respectively. Also the relative capacity and efficiency of rectifier output were considered at 75% and 85%, respectively.

Table 3. Maximum and minimum wind speed for each province.

Region (province)	Maximum wind speed (m/s)	Minimum wind speed (m/s)
Bouira	4.69	1.8
Boumerdes	7.08	2.55
Media	5.58	1.86

Table 4. Battery properties.

Kinetic battery model	
Nominal voltage (V)	6
Nominal capacity (kWh)	2.37
Maximum capacity (Ah)	396
Capacity ratio	0.242
Rate constant (1/hr)	1.77
Roundtrip efficiency (%)	85
Max charge current (A)	18
Max charge rate (A/Ah)	1

4 CALCULATION METHOD OF HYBRID HYDRO/WIND SYSTEM

4.1 Wind power output

The wind power output is obtained by HOMER, which follows these three steps:

4.1.1 Calculating hub height wind speed

$$U_{\text{hub}} = U_{\text{anem}} \frac{\ln(Z_{\text{hub}}/Z_0)}{\ln(Z_{\text{anem}}/Z_0)} \quad (1)$$

$$U_{\text{hub}} = U_{\text{anem}} \left(\frac{Z_{\text{hub}}}{Z_{\text{anem}}} \right)^{\alpha} \quad (2)$$

Eq. (1) is used by HOMER to calculate the hub height wind speed applying the logarithmic law, while Eq. (2) is used by HOMER to calculate the hub height wind speed applying the power law.

4.1.2 Calculating turbine power output at standard air density

The second step is that HOMER determines the power output from the wind turbine's power curve at hub height wind speed under conditions of standard temperature and pressure (STP).

4.1.3 Applying density correction

To adjust the wind turbine performance from STP to actual conditions, HOMER uses the air density ratio, according to Eq. (3):

$$P_{\text{WTG}} = \left(\frac{\rho}{\rho_0} \right) P_{\text{WTG,STP}} \quad (3)$$

4.2 The hydro power output

Equation (4) is used by HOMER:

$$P_{\text{hyd}} = \frac{\eta_{\text{hyd}} \cdot \rho_{\text{water}} \cdot g \cdot h_{\text{net}} \cdot Q_{\text{turbine}}}{1000 \text{ W/kW}} \quad (4)$$

4.3 Total annualized cost

Equation (5) is used by HOMER, the unit used is \$/year:

$$C_{\text{ann,tot}} = \text{CRF}(i, R_{\text{proj}}) \cdot C_{\text{NPC,tot}} \quad (5)$$

HOMER uses the total annualized cost to calculate the leveled cost of energy.

5 RESULTS AND DISCUSSION

• Case 1: water flow rate of 10 m³/s

Figure 6 shows the output power of the wind turbine (in terms of hour) throughout the year for case 1, while Figure 7 shows the state of charge for all days of the year. The state of charge is almost 100% for most days of autumn, winter and spring, whereas in summer this value varies from 30% to 100% according to the hours of the day.

Tables 5 and 6 provide various costs of the different components of wind turbine (case 1) for wind speeds of 1.8 m/s and 4.69 m/s, respectively, where water flow is considered 10 m³/s.

Figures 8 and 9 demonstrate the monthly electric production for case 1 with different wind speeds. It is noticed that the peak appears in December and April with the maximum production in autumn and winter and the minimum in summer. In this case, it appears that the hydro power that ranges between 3.927 and 5.543 kWh is not sufficient to meet the required load in this area. However, the amount of electricity generated by the hybrid hydro/wind system is sufficient to cover the required load in this area.

Table 7 gives a comparison based on the total financial costs between the two systems for water flow of 10 m³/s. According to this table, the minimum cost is 400 082 USD, which is related to a wind speed of 4.69 m/s. Conversely, the maximum cost is 1 669 921.53 USD, which is related to a wind speed of 1.8 m/s.

• Case 2: water flow rate of 19 m³/s

Figure 10 shows the output power of the wind turbine (in terms of hour) throughout the year for case 2, while Figure 11 shows the state of charge percentage during all the year for case 2, where the ratio of the state of charge is recorded at 100% in the afternoon, for all days of the year.

Various costs of each component of the wind turbine with the wind speed of 2.55 m/s and 7.08 m/s for case 2 are provided in Tables 8 and 9, respectively. Figures 12 and 13 demonstrate the monthly electric energy production for case 2 with different wind

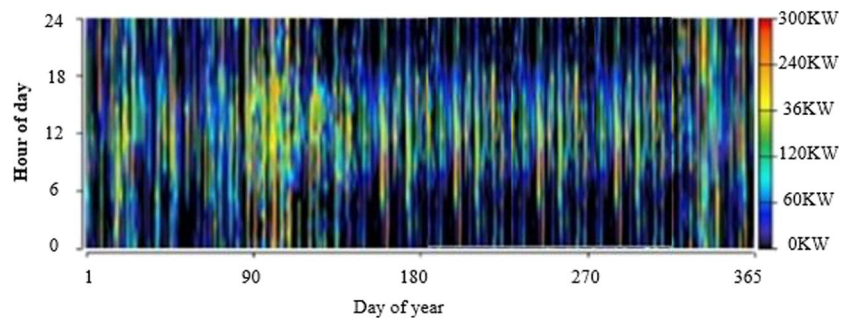


Figure 6. Wind turbine output power for case 1.

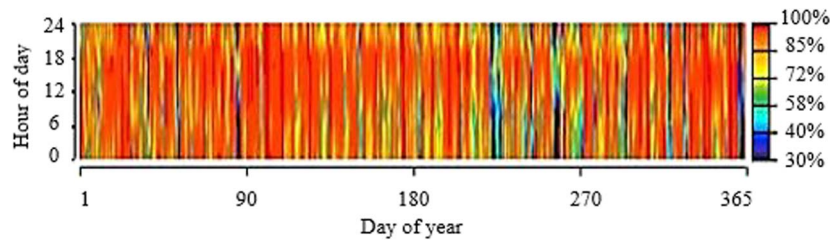


Figure 7. State of charge in case 1.

Table 5. Costs at wind speed of 1.8 m/s for case 1.

Component	Capital (USD)	Replacement (USD)	Operations & maintenance (USD)	Salvage (USD)	Total cost (USD)
Converter	369.64	11.78	238.93	6.64	613.71
Generic 10 kW	1300000.00	414449.56	168057.72	233569.90	1648938.37
Torjan L16P	7770.00	11275.01	1913.72	588.83	20369.45
Hybrid system	1308139.64	425736.35	170209.92	234164.38	1669921.53

Table 6. Costs at wind speed of 4.69 m/s for case 1.

Component	Capital (USD)	Replacement (USD)	Operations & maintenance (USD)	Salvage (USD)	Total cost (USD)
Converter	325.46	10.38	210.37	5.85	540.35
Generic 10 kW	300000.00	95642.21	38782.55	53900.52	380424.24
Torjan L16P	7350.00	10532.92	1809.85	674.70	19018.08
Hybrid system	307655.46	106185.50	40802.77	54581.06	400082.67

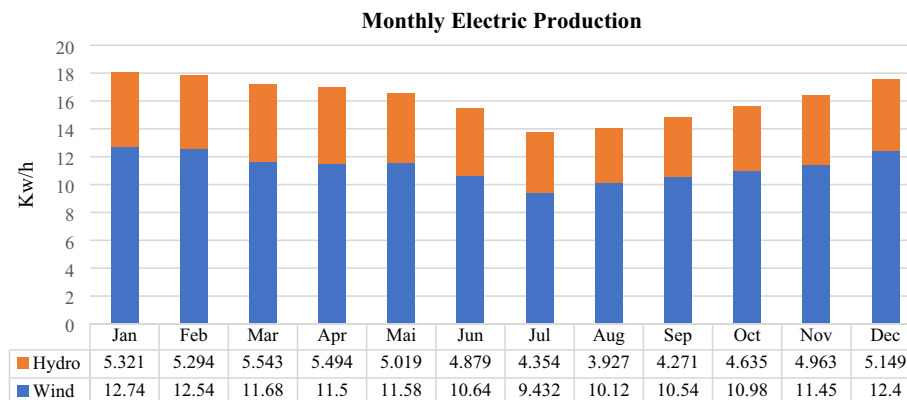


Figure 8. Monthly wind turbine electric production for case 1 with wind speed 1.8 m/s.

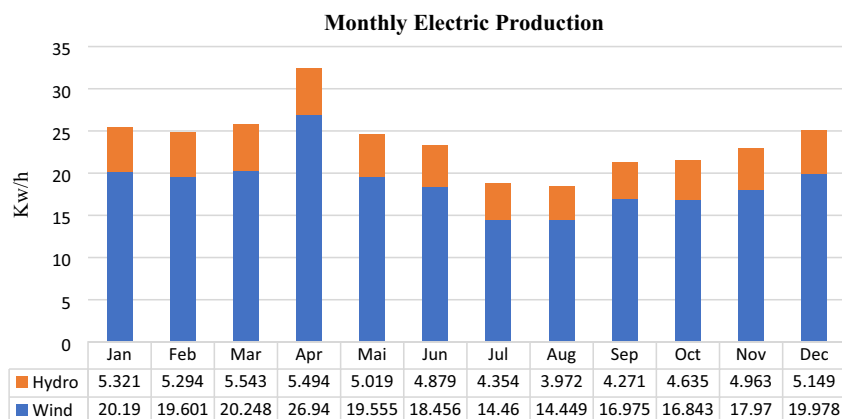


Figure 9. Monthly wind turbine electric production for case 1 with wind speed 4.69 m/s.

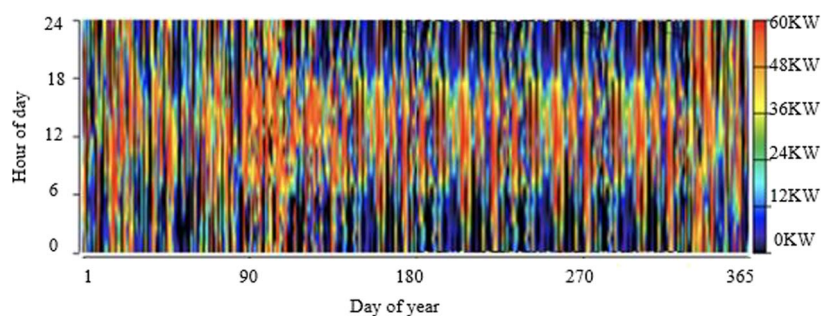


Figure 10. Wind turbine power output for case 2.

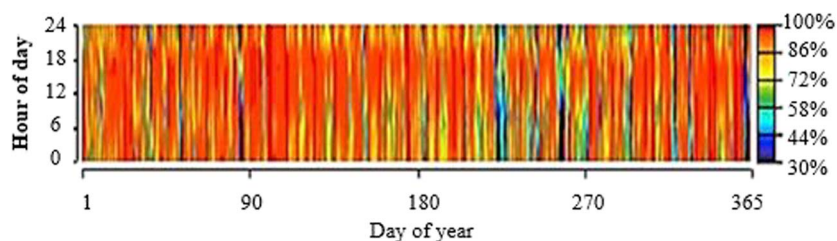


Figure 11. State of charge for case 2.

Table 7. Total cost for each hybrid system for case 1 with different wind speed.

Wind speed (m/s)	Total cost (USD)
1.8	1 669 921.53
4.69	400 082.67

speeds. It is revealed that the maximum electricity production occurs from December to June, while the minimum electricity production occurs from July to November. In the second case, in winter and spring seasons, the amount of electricity produced by the hydro system exceeds the value of the required load in this region, but in the summer and autumn seasons, the amount of electricity produced is insufficient.

For the hybrid system (hydro/wind), the amount of electricity produced is sufficient to cover the required load all throughout the year.

A financial costs comparison between two systems for 19 m³/s water flow is shown in Table 10, which confirms the cost reduction with wind speed increment.

• **Case 3: water flow rate of 25.3 m³/s**

Figure 14 shows the state of charge for all days of the year for case 3. It is clear that the state of charge is 100% in all hours of the day during winter, autumn and spring, while its value lies within the range of 40%–90% in summer.

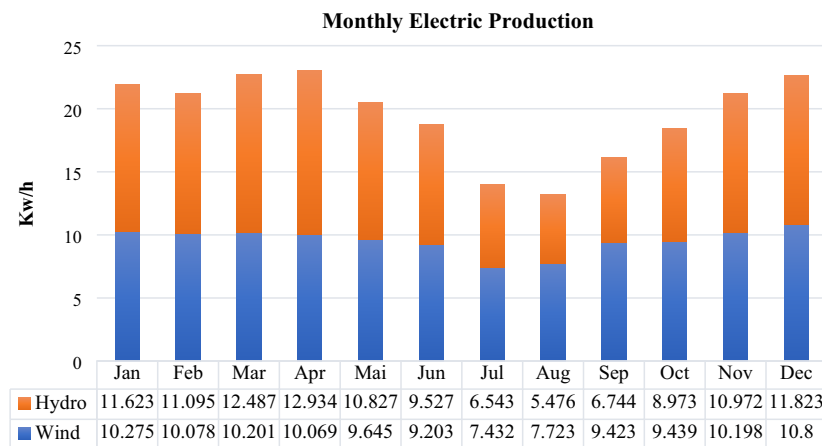
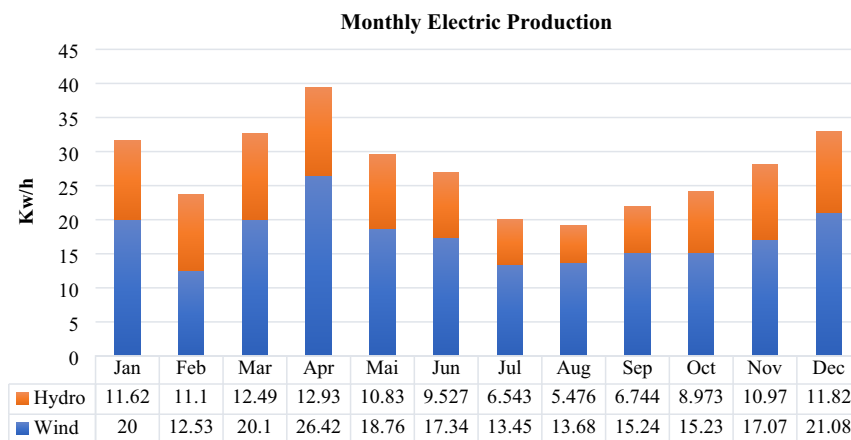
Figure 15 demonstrates the monthly hydro electricity production for case 3 (25.3 m³/s water flow). The electric power production is approximately fixed from December to June with a

Table 8. Costs at wind speed of 2.55 m/s for case 2.

Component	Capital (USD)	Replacement (USD)	Operations & maintenance (USD)	Salvage (USD)	Total cost (USD)
Converter	299.79.50	9.90	200.70	5.58	496.06
Generic 10 kW	950023.00	302867.08	122791.41	170684.97	1204993.42
Torjan L16P	57770.00	6864.31	32059.79	2527.73	76947.36
System	1008068.79	309740.82	137845.32	173218.07	1282436.86

Table 9. Costs at wind speed of 7.08 m/s for case 2.

Component	Capital (USD)	Replacement (USD)	Operations & maintenance (USD)	Salvage (USD)	Total cost (USD)
Converter	310.50	9.90	200.70	5.58	515.52
Generic 10 kW	20000.00	63761.47	25855.03	35933.68	253682.83
Trojan L16P	57770.00	6864.31	32059.79	2527.73	76947.36
System	258080.50	70635.68	40896.52	38466.99	331145.71

**Figure 12.** Monthly hydro–wind electric production for case 2 with wind speed 2.55 m/s.**Figure 13.** Monthly hydro–wind electric production for case 2 with wind speed 7.08 m/s.

slight decrease in some months of summer, and the minimum production is obtained in August and September.

Various costs related to case 3 are shown in Table 11 where the system cost is the same as the hydroelectric, because the amount of electricity generated by the water flow in this case is sufficient

to provide the required electrical load and there is no need for another system such as wind generator.

Figure 16 shows the cost changes based on wind speed. According to this graph, for all cases the greater the wind speed the lower the cost of the system. Comparison of costs shows

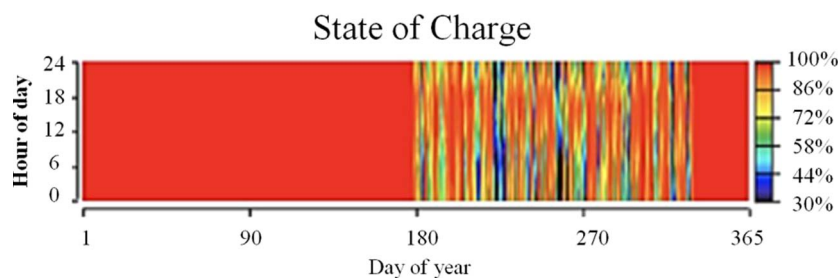


Figure 14. State of charge for case 3.

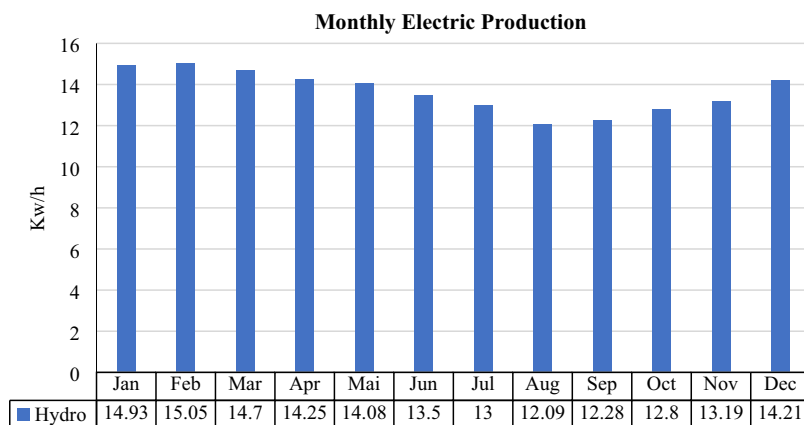


Figure 15. Monthly hydro-electric production for case 3.

Table 10. Total financial cost for each case at water flow rate of 19 m³/s.

Wind speed (m/s)	Total cost (USD)
2.55	1 282 436.86
7.08	331 145.71

that case 3 has the minimum cost among the three considered systems.

In the first case, when the average water flow rate is 10 m³/s it can be found that the cost of producing the required energy decreases as the wind speed increases. The system cost decreases from 1 669 921 USD (wind speed, 1.8 m/s) to 40008.27 USD (wind speed, 4.69 m/s) when water flow rate is 10 m³/s. However, the cost values for water flow rate of 19 m³/s decreases from 1 282 437 USD (wind speed, 2.55 m/s) to 331 146 USD (wind speed, 7.08 m/s). It can be inferred from results that a significant decrement occurred in the system costs when water flow rate is considered at 25.3 m³/s so that it is decreased to the value of 61 330.46 USD, which is the most cost-effective case.

6 CONCLUSIONS

This research paper gave an overall study of a wind–hydro power system with three different cases, and each case has different

values of water flow rates and wind speeds. This paper contributes to the literature by focusing on the renewable energy generators and the advantage of the hybridization of these resources. The developed system can be helpful for many practicing engineers in their future studies and works in building hybrid energy generators such as wind–hydro energy system while making sure to be able to feed the load demand in the studied area.

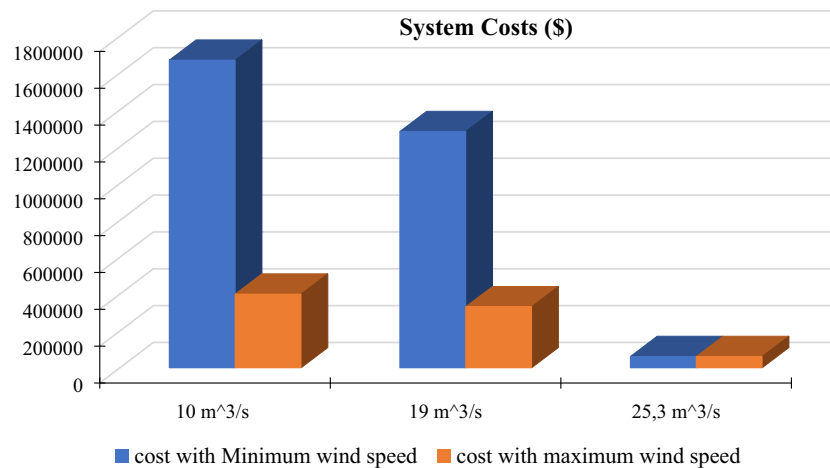
The water flow speed in the main branch of Yesser valley is estimated to be 24.6 m³/s, where other regions may reach only 10 m³/s. The possibility of constructing a hybrid wind/hydro system was discussed in order to provide the amount of load needed in some of the valley's areas.

A good electricity production from hydro in the third case was shown with a maximum power generation during the winter season with the lower cost within the three cases of 61 330.46 USD. For the second case a fluctuation in power generation was observed, with a lack of energy in the summer season. To compensate this, a second power generator (wind generator) has been added, and a hydro/wind hybrid system has been constructed where the cost for 2.55 m/s wind speed was 1 282 436 USD and for 7.08 m/s wind speed was 331 146 USD. For the first case the hydro power was not sufficient to feed the load unless a wind generator was added with a minimum cost of 400 082 USD observed when the wind speed was 4.69 m/s and a maximum cost of 1 669 921 USD when the wind speed was 1.8 m/s.

By assuming that the climate has no extra ordinary change, the study proves that the mentioned hybrid system provided more

Table 11. Costs in USD for case 3 (25.3 m³/s).

Component	Initial capital (USD)	Operations & maintenance (USD)	Salvage (USD)	Total cost (USD)
Hydro-electric	50,000.0	12,927.52	1,597.05	61,330.46
System	50,000.0	12,927.52	1,597.05	61,330.46

**Figure 16.** The cost changes according to wind speed.

than the necessary electrical power to the different loads for the other branches of Yesser valley, and we also propose to transfer the surplus energy to other branches. This hybrid system is economic and environmentally friendly with a much better capacity than the individual one, which made it efficient enough to satisfy the power demand in Yesser valley.

REFERENCES

- [1] Zou C, Zhao Q, Zhang G *et al.* Energy revolution: from a fossil energy era to a new energy era. *Nat Gas Ind B* 2016;**3**:1–11.
- [2] Hardisty PE, Clark TS, Hynes RG. Life cycle greenhouse gas emissions from electricity generation: a comparative analysis of Australian energy sources. *Energies* 2012;**5**:872–97.
- [3] Jalili M, Chitsaz A, Hashemian M *et al.* Economic and environmental assessment using emergy of a geothermal power plant. *Energy Convers Manag* 2021;**228**:113666.
- [4] Jafary S, Khalilarya S, Shawabkeh A *et al.* A complete energetic and exergetic analysis of a solar powered trigeneration system with two novel organic Rankine cycle (ORC) configurations. *J Clean Prod* 2021;**281**:124552.
- [5] Liu C, Hashemian M, Shawabkeh A *et al.* CFD-based irreversibility analysis of avant-garde semi-O/O-shape grooving fashions of solar pond heat trade-off unit. *Renew Energy* 2021;**171**:328–43.
- [6] Cao Y, Ayed H, Hashemian M *et al.* Thermal/frictional performance of spiral pipe with ring-shape depression used as in-pond heat exchanger. *Sol Energy* 2021;**224**:742–56.
- [7] Elhadidy MA. Performance evaluation of hybrid (wind/solar/diesel) power systems. *Renew Energy* 2002;**26**:401–13.
- [8] Elhadidy MA, Shaahid SM. Decentralized/stand-alone hybrid wind-diesel power systems to meet residential loads of hot coastal regions. *Energy Convers Manag* 2005;**46**:2501–13.
- [9] Shaahid SM, Elhadidy MA. Prospect of autonomous/ stand-alone hybrid (photovoltaic+diesel+battery) power systems in commercial applications in hot regions. *Renew Energy* 2004;**29**:165–77.
- [10] Bagul AD, Salameh ZM, Borowy B. Sizing of a stand-alone hybrid wind photovoltaic system using a three-event density approximation. *Sol Energy* 1996;**56**:323–35.
- [11] Kaldellis JK, Kondili E, Filios A. Sizing a hybrid wind-diesel stand-alone system on the basis of minimum long-term electricity production cost. *Appl Energy* 2006;**83**:1384–403.
- [12] Yoo K, Park E, Kim H *et al.* Optimized renewable and sustainable electricity generation systems for Ulleungdo Island in South Korea. *Sustainability* 2014;**6**:7883–93.
- [13] Melzi B, Kefif N, El Haj Assad M *et al.* Modelling and optimal design of hybrid power system photovoltaic/solid oxide fuel cell for a Mediterranean City. *Energy Eng* 2021;**118**:1767–1781.
- [14] Merriam Webster Dictionary. <https://www.merriam-webster.com/dictionary/efficiency> (27 October 2019, date last accessed).
- [15] Wang CN, Nguyen TD, Yu MC. Energy use efficiency past-to-future evaluation: an international comparison. *Energies* 2019;**12**:3804.
- [16] Ramírez Díaz A, Ramos-Real FJ, Marrero GA *et al.* Impact of electric vehicles as distributed energy storage in isolated systems: the case of Tenerife. *Sustainability* 2015;**7**:15152–78.
- [17] Asif M, Muneer T. Energy supply, its demand and security issues for developed and emerging economies. *Renew Sust Energy Rev* 2007;**11**:1388–413.
- [18] Deshmukha MK, Deshmukh SS. Modeling of hybrid renewable energy systems. *Renew Sust Energy Rev* 2008;**12**:235–49.
- [19] Onder O, Leyla O. Exergy and reliability analysis of wind turbine systems: a case study. *Renew Sust Energy Rev* 2007;**11**:1811–26.
- [20] Nand K, Saini RP, Singh SP. A review on hydropower plant models and control. *Renew Sust Energy Rev* 2007;**11**:776–96.
- [21] Shivraht Y, Patel BN, Thirumalasetty S *et al.* Design & integration of wind-solar hybrid energy system for drip irrigation pumping application. *International Journal of Modern Engineering Research (IJMER)* 2012;**2**:2947–2950.
- [22] Maheshwari Z, Ramakumar R. Smart Integrated Renewable Energy Systems (SIREs): a novel approach for sustainable development. *Energies* 2017;**10**:1145.

- [23] Garcia Latorre FJ, Quintana JJ, de la Nuez I. Technical and economic evaluation of the integration of a wind-hydro system in El Hierro island. *Renew Energy* 2018;**134**:186–193.
- [24] Li H, Mahmud MA, Arzaghi E *et al.* Assessments of economic benefits for hydro-wind power systems: development of advanced model and quantitative method for reducing the power wastage. *J Clean Prod* 2020;123823. <https://doi.org/10.1016/j.jclepro.2020.123823>.
- [25] Eliseu A, Castro R. Wind-hydro hybrid park operating strategies. *Sust Energy Technol Assess* 2019;**36**:100561.
- [26] Xie T, Zhang C, Wang T *et al.* Optimization and service lifetime prediction of hydro-wind power complementary system. *J Clean Prod* 2021;**291**:125983.
- [27] Canales FA, Beluco A, Mendes CAB. A comparative study of a wind hydro hybrid system with water storage capacity: conventional reservoir or pumped storage plant? *J Energy Storage* 2015;**4**:96–105.
- [28] National Renewable Energy Laboratory, HOMER Getting Started Guide Version 2.1, NREL, 2005.
- [29] NASA. www.nasa.gov.
- [30] Djaouida B, Ourida K, Mebarki M *et al.* The Odonata of Wadi Isser (Kabylia, Algeria): status and environmental determinants of their distribution. *Revue d'Ecologie* 2015;**70**:248–260.
- [31] World Weather Online. www.worldweatheronline.com